

CO(i)NQUEST

A gamified expense tracker with a local AI coach — log, level up, and conquer the month

ONE-PAGE SOFTWARE SPEC v1.0 · 12 JUL 2026

AI Engineering Summer Training 2026

Week 2 – AI-assisted development · Day 1



01 THE PROBLEM IT SOLVES

Expense trackers have a retention problem: logging what you spend is a chore with no feedback loop, so most people quit within a week — and students rarely find out where the allowance went. **CoinQuest reframes the habit as a game:** the monthly budget is a health bar, consistent logging builds streaks and XP, and a local AI coach turns the numbers into advice — or a roast. Tracking money should feel like maintaining a save file, not filing taxes.

02 KEY FEATURES

COMMON · CORE SLICE

built day 2 – one thin path through every layer

Log an expense

amount, category, note, date → POST /api/expenses → SQLite

Expense list

newest first from `GET /api/expenses` — proof the full stack is wired

RARE · V1.0

shipped by day 5

Budget HP bar

drains in real time; green → yellow → red, and 0 HP triggers a "defeated" state

Streak & XP

consecutive logging days keep the streak alive; every entry earns XP

Monthly summary

backend computes totals at `GET /api/summary` ; Chart.js draws the charts

AI coach

Ollama turns pre-computed stats into a review + one tip — Coach / Roast / Pirate modes

Monthly boss fight

top spending category becomes the month's boss – "Restaurant Monster dealt 720 SAR"

EPIC · STRETCH

only if day 5 has hours left

If time allows

AI expense entry ("Lunch at AlBaik 28 SAR" → structured JSON) · AI weekly quests · achievements & titles — full detail in [SPEC.md](#)

TECH LOADOUT

FastAPI · Python 3

SQLite

HTML · CSS · JS

Chart.js

Ollama local LLM

Git + GitHub · tags v0.1 → v1.0

03 CONSTRAINTS – WHAT IT WILL NOT DO

- ✗ **No accounts, logins, or multi-user.** One player, on localhost.
- ✗ **No bank sync, card data, or real financial integrations.**
- ✗ **No cloud deployment.** Data never leaves the machine.
- ✗ **The AI never does arithmetic.** Python computes every number; the model only writes text.
- ✗ **AI output is advisory only.** It never creates, edits, or deletes records.
- ✗ **One person + one AI assistant, five days.** Features that threaten the deadline move down a tier.

DEFINITION OF DONE

- ☐ Clone, run one command, add an expense, watch the HP bar drop.
- ☐ Monthly summary shows real charted totals plus the AI coach's boss report.
- ☐ Repo history tells the week's story: daily tags v0.1 → v1.0, `REVIEW.md` , `BUGLOG.md` .